Hamza Cherkaoui

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Interests

Convex Optimization; Dictionary Learning; Neural Networks; Total Variation regularization; Semi blind deconvolution; Clinical Investigation; Pharmacology

Research experience

2017–2021 Phd student

PhD program, PARIETAL research team, CEA-Saclay / INRIA-Saclay, Paris, France

- *Research subject: 'Efficient whole brain estimation of the haemodynamic response function for TV-regularized semi-blind deconvolution of neural activity in fMRI'*,
- *Supervisors:* Dr. Philippe Ciuciu, Dr. Thomas Moreau and Dr. Claire Leroy
- Part of a team involving doctors and mathematicians for developping algorithmic tools to investigate the brain in a healthy or a disease context,
- *Convex Optimization, Dictionary Learning, Neural Networks, Total Variation, Deconvolution.*
- Published 5 International Conference Proceedings and 1 Journal Articles.

Education

2016–2017

Master program, Ecole Polytechnique, Palaiseau, France

- *Master of Research:* Data Sciences
- Machine Learning Theory, Kernel methods, Probabilistic Graphical Models, Convex Optimization.

2015–2016

Master program, University Lille 1, Villeneuve d'Ascq, France

- *Master of Research:* major in applied mathematics
- focused on Probability and Statistic.

2012–2016

Ecole Centrale de Lille, Villeneuve d'Ascq, France

- *Master of Engineering:* diplome d'ingénieur de l'Ecole Centrale de Lille.
- Cross disciplinary courses, centered around mathematics and computer science.

Other Experiences

2021 August–November

Vacancy research engineer, BioMaps research team, SHFJ, CEA-Saclay, Orsay, France

- I continue my research on the estimation of the Haemodynamic Response Function (HRF) from fMRI data in a pharmacological context.

2017 April–September

Researcher assistant (internship), PARIETAL research team, Neurospin laboratory CEA-Saclay, France

- I searched to optimize the reconstruction process of MRI image, with Compressed Sensing tools, to enhance image quality.

2016 June–August

Research engineer (internship), MODAL research team, INRIA Lille, France

- *Software development project:* Development of a new R package providing tools based on a efficient `C++` implementation to cluster categorical functions.

2015 April–July

Data core-developper (internship), DataRobot, Boston, USA

- *Software development project:* Implementation of the decision tree induction algorithms (from `Scikit-Learn`) used in the company to support multi-threaded training.

2014 May–December

Software developper (internship), Logilab, Paris, France

- *Software development project:* Implementation of a Python library based on `CGAL` library (`Python/C++ binding`) & development of a web-client toolkit to use a computation platform.

Languages & Computer Science skills

- *French:* mother-tongue
- *English:* (TOEIC 970/990) fluent
- *Collaborative tools:* `Git, Mercurial`
- *Languages:* `Python`, `C++, C, R`
- *Libraries:* `Numpy, Pandas, Numba, pyTorch, Scipy, Scikit-learn, Nilearn`

Publications

[1] Cherkaoui, H. and Moreau, T. and Halimi, A. and Leroy, C. and Ciuciu, P. *Multivariate semi-blind deconvolution of fMRI time series*, NeuroImage, 2021.

[2] Cherkaoui, H. *Efficient whole brain estimation of the haemodynamic response function for TV-regularized semi-blind deconvolution of neural activity in fMRI*, Ph.D. thesis, 2021.

[3] Cherkaoui, H. and Moreau, T. and Ciuciu, P. and Fernandez, B. and Bottlaender, M. and Tournier, N. and Leroy, C. *Characterization of the haemodynamic response function after a buprenorphine challenge study in Human healthy volunteer*, 27th Annual Meeting and Educational Courses of OHBM, 2021.

[4] Cherkaoui, H. and Sulam, J. and Moreau, T. *Learning to solve TV regularised problems with unrolled algorithms*, 34th Conference and Workshop on Neural Information Processing Systems (NeurIPS), 2020.

[5] Cherkaoui, H. and Moreau, T. and Halimi, A. and Ciuciu, P. *Sparsity-based blind deconvolution of neural activation signal in fMRI*, 44th International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019.

[6] Cherkaoui, H. and Moreau, T. and Halimi, A. and Ciuciu, P. *fMRI BOLD signal decomposition using a multivariate low-rank model*, 27th European Signal Processing Conference (EUSIPCO), 2019.

[7] Cherkaoui, H. and Gueddari, L. and Lazarus, C. and Grigis, A. and Poupon, F. and Vignaud, A. and Farrens, S. and Starck, J.-L.. and Ciuciu, P. *Analysis vs synthesis-based regularization for combined compressed sensing and parallel MRI reconstruction at 7 tesla*, 26th European Signal Processing Conference (EUSIPCO), 2018.

Developped Open Source Python packages

2020 **Carpet**, Fast-minimization of Total Variation regularized problems. carpet
- Minimize 1d TV regularized problems with neural network.

2019 **HemoLearn**, Multivariate estimation of the HRF from fMRI data. hemolearn
- Disentangle the neurovascular coupling from the neural activation signal in a multivariate fashion for fMRI data.

2018 **pyBOLD**, Univariate estimation of the HRF from fMRI data. pybold
- Disentangle the neurovascular coupling from the neural activation signal in a univariate fashion for fMRI data.

2018 **pysap**, MRI image reconstruction. pysap
- MRI image reconstruction based on Compressed Sensing (CS).